Tanmay Bhuskute

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SUMMARY

Software Engineer and M.S. Computer Science candidate (Dec 2026) building large-scale backend services and CI/CD automation with Java, Spring, Hibernate, REST APIs, AWS, Docker, and Kubernetes. Experienced with Jenkins, SonarQube, Terraform, PostgreSQL, MySQL, and production monitoring/on-call support.

EDUCATION

Master of Science in Computer Science Aug 2024 - Dec 2026
University of Cincinnati
Relevant Coursework: Advanced Algorithms, Deep Learning, Intelligent Data Analysis, Database Theory.

Bachelor of Engineering in Computer Engineering Aug 2018 - Jul 2022
AISSMS College of Engineering
Relevant Coursework: Data Mining, Data Analytics, Machine Learning, Object-Oriented Programming.

TECHNICAL SKILLS

Programming: Java, Python, JavaScript, TypeScript, C/C++, SQL, Go.

Backend & Frameworks: Spring, Hibernate, REST APIs, Node.js, Django, Flask, FastAPI.

Databases: PostgreSQL, MySQL, Oracle, Redis.

CI/CD & Cloud: Git, Jenkins, Maven, Gradle, SonarQube, AWS, Docker, Kubernetes, Terraform, JFrog Artifactory.

Testing & Monitoring: JUnit, Splunk, Datadog, Postman, Linux, Bash.

WORK EXPERIENCE

Graduate Researcher —- University of Cincinnati Feb 2026 – Present

- Designed two-stage memory-augmented LLM pipeline decoupling video understanding from device automation for Android GUI bug reproduction, evaluated against **ViBR** (GPT-4o + GroundingDINO + CLIP baseline) across 53 runs on 20+ apps.
- Demonstrated **25.7x latency reduction** (39.4s vs 1012.1s avg per run) while preserving task-completion fidelity.

Software Development Engineer —- Siemens Industry Software Jul 2022 – Jun 2024

- Built and maintained backend infrastructure automation in Python with REST API integrations, replacing legacy Ruby scripts and reducing deployment time by **40%** across AWS environments.
- Executed **44 production environment upgrades** in 4 months, diagnosing live infrastructure issues across Linux, EC2, databases, and enterprise application services with on-call incident response.
- Designed **Ansible-based Blue-Green deployment** framework using Jenkins CI/CD pipelines, reducing production upgrade downtime by **50%**.
- Migrated **40+ production environments** to Kubernetes using ArgoCD, Helm, GitLab CI/CD, Terraform, and HashiCorp Vault for secrets management.
- Served as Quality Coach and SonarQube Specialist, increasing automated code coverage from **35% to 78%**; mentored **5 software engineers** through structured onboarding.

Software Developer Intern —- Siemens Industry Software Mar 2022 – Jun 2022

- Automated provisioning of AWS resources with **Terraform**, reducing deployment time by **30%**.
- Developed **Go + Terratest** framework, catching **95%** of misconfigurations pre-production.

Software Developer Intern —- Government of Maharashtra, Water Resource Department Sep 2021 – Dec 2021

- Built and deployed a full-stack **Java-backed and MySQL** workflow application that digitized departmental record management, automated manual processes, and improved operational efficiency by **40%**.

PROJECTS AND RESEARCH

NYC Mobility Forecasting [Python, PostgreSQL, Redis, Docker, FastAPI] Jul 2026

- Built ML system forecasting NYC taxi/rideshare demand across **263 zones** (**2.3M-row** hourly dataset); designed two-stage Poisson hurdle model improving MAPE **18.77% to 16.52%** and cutting RMSE **43%** over baseline Random Forest.
- Built RESTful API service integrated with PostgreSQL and Redis caching, containerized with Docker for scalable deployment.

EXTRACURRICULAR ACTIVITIES

- **Finalist, RevolutionUC 2025 Hackathon** (300 participants, 31 schools) | **SAFe Practitioner 5.1** (Mar 2023).